① $G = \{\{S, C\}, \{a, b\}, P, S\}$.

$S \to aCa$

$C \to aCa \mid b$.

$L(G) = ?$

$G = \{aba, aabaa, aaabaaa, \ldots\}$ → palindrome

So, $L(G) = a^n b a^n$.

---

② $S \to aAS \mid a$

$A \to SbA \mid SS \mid ba$.

Prove $S \to aabbaa$

Derivation tree

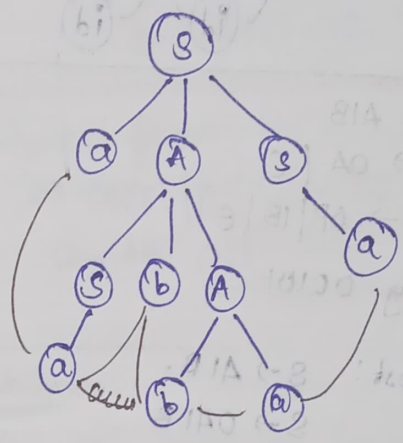
$S \to aAS$.

$S \to aSbAS$.

$a \to aabAS$.

$a \to aabbaS$.

$a \to aabbaa$.



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③ $G = (\{S\}, \{0, 1\}, \{S \to 0S1, S \to \epsilon\}, S)$

$L(G) = ?$

$S \to 0S1 \mid \epsilon$.

$L(G) = \{01, 0011, 000111, \ldots\}$.

$L(G) = \{0^n 1^n \mid n \geq 0\}$

---

④ Derivation tree: It is graphical representation for derivation of given production rule for CFG.

---

⑤ $L = \{a^n b^n \mid n \geq 1\}$, $L(G) = ?$

$L(G) = \{ab, aabb, aaabbb, \ldots\}$

$S \to aSb \mid ab$.

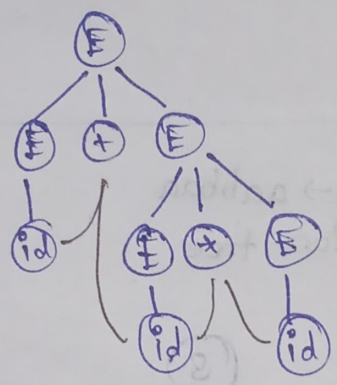
⑥ $E \to E+E \mid E*E \mid (E) \mid id.$

Show that id + id * id.

$E \to E+E$

$E \to E+E*E$

$E \to id+id*id.$



⑦ $S \to A \mid B$

$A \to 0A \mid \epsilon$

$B \to 0B \mid 1B \mid \epsilon$

string 00101.

Left most:

$S \to A \mid B.$

$S \to 0A \mid B$

$S \to 00A \mid B$

$S \to 00A \mid B.$

$S \to 001B.$

$S \to 0010B$

$S \to 00101 \mid B$

$S \to 00101 \mid \epsilon$

$S \to 00101$

Right most:

$S \to A \mid B$

$S \to A \mid 0B$

$S \to A \mid 0 \mid B$

$S \to A \mid 01$

$S \to 0A \mid 01$

$S \to 00A \mid 01$

$S \to 0010 \mid$

⑧ $\Sigma = \{a, b, (,), +, *, \ldots \epsilon\}$

Construct all Σ in CFG.

$S \to S+S$

$S \to S*S$

$S \to (S)$

$S \to a \mid b.$

$S \to \epsilon.$

$S \to S+S \mid S*S \mid (S) \mid a \mid b \mid \epsilon.$

⑨ $S \to 0B \mid 1A.$

$A \to 0 \mid 0S \mid 1AA$

$B \to 1 \mid 1S \mid 0BB.$

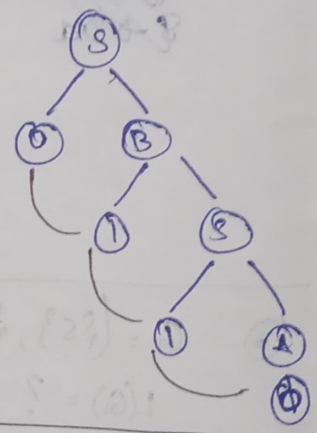
string 0110. ⇒ Rightmost

$S \to 0B.$

$S \to 01S$

$S \to 011A$

$S \to 0110.$



⑩ $S \to aB \mid bA.$

$A \to aS \mid bAA \mid a$

$B \to bS \mid aBB \mid b.$

String aaabbabbba ⇒ Left most.

$S \to aB$

$S \to aaBB$

$S \to aaaBBB.$

$S \to aaabSBB$

$S \to aaabbABB$

$S \to aaabbaSBB$

$S \to aaabbabABB$

$S \to aaabbab$

(10) $S \to aB \mid bA$
$A \to aS \mid bAA \mid a$.
$B \to bS \mid aBB \mid b$.

string $\Rightarrow$ aaabbabbba
↳ left most.

$S \to aB$
$S \to aaBB$
$S \to aaaBBB$.
$S \to aaabBB$
~~$S \to aaabbB$~~.
~~$S \to aaabbaBB$~~
~~$S \to aaabbabB$~~
~~$S \to aaabbabbAA$~~
~~$S \to aaabbabbbAA$~~.
~~$S \to aaabbSB$~~.
~~$S \to aaabbaBB$~~
~~$S \to aaabbabSB$~~
~~$S \to aaabbab$~~

aaabbSB
aaabbaBB
aaabbabB
aaabbabbS
aaabbabbbA
aaabbabbba.

(11) $\Sigma = \{a, b, (,), +, *, \ldots \epsilon\}$.
CFG?

$S \to 0S \mid 1S \mid \epsilon$.

(12) $S \to a \mid Sa \mid bSS \mid SSb \mid SbS$

ababaa..
↓
Two trees
Ambiguous

(13) $S \to aSb \mid aAb$
$A \to bAa \mid ba$.

aabbabb

$L = \{a^m b^n a^n b^m \mid n, m \geq 1\}$

(14) CFG $\{a, b\}$ equal no. of a's & b's.

$V = \{S, A, B\}$

$S \to aSb \mid ab$.
(or)
$S \to aB \mid bA$
$A \to a \mid aS \mid bAA$.
$B \to b \mid bS \mid aBB$.

(15) Ambiguous from
$S \to SS(S) \mid (S) \mid S(S)S \mid \epsilon$.
Consider ()()
↳ Two trees.

(16) Ambiguous:
For deriving the same input string if there is more than one parse tree, then the grammer is said to be ambiguous

(17) $S \to aCa$
$C \to aCa \mid b$.

aCa
aaCaa
aabaa

$L(G) = \{a^n b a^n \mid n \geq 1\}$.

(18) $L = \{a^n b a^m \mid n, m \geq 1\}$

$S \to aCaa$.

$C \to aCa \mid b$.

aCaa

aaCaaa

aabaaa.

(19) Unambigous grammer:

It has a unique derivation tree for a given string.

(20) Parse Tree:

The Parse tree is the visual representation of the derivation of unit production rules for a CFG

(21) Uses of CFG:

It is used to check the syntax of the source language statement

(22) $S \to aB \mid bA$.

$A \to a \mid aS \mid bAA$

$B \to b \mid bS \mid aBB$.

String ~~aaa bbbbbba~~.

Find a) LMD, b) RMD.

LMD ⇒ $S \to aB$

$S \to aaS$

$S \to aaaB$

$S \to aaabS$

$S \to aaabbA$

$S \to aaabbaB$

$S \to aaabbabS$

$S \to aaabbabbA$

$S \to aB$

$S \to aaBB$

$S \to aaaBBB$

$S \to aaabBB$

$S \to aaabbB$

$S \to aaabbaBB$

$S \to aaabbabB$

$S \to aaabbabbS$

$S \to aaabbabbbA$

$S \to aaabbabbba$

(23) $(0+1)^* \Rightarrow$ CFG = ?

~~L(G) = {0S1}~~

$S \to 0S0 \mid 1S1 \mid \epsilon$.

(24) Equal no. of a's & b's ⇒ CFG.

$S \to aB \mid bA$.

$A \to a \mid aS \mid bAA$.

$B \to b \mid bS \mid aBB$.

$G = V, T, P, S$

$V = \{S, A, B\}$

$T = \{a, b\}$

$L(G) = \{a^n b^n \mid n \geq 1\}$

(25) Doubt

(26) Push Down Automata:

The pushdown Automata (PDA) Consist of seven components,

1. Finite no. of states (Q).
2. Alphabet (Σ)
3. Initial state (Q_0)
4. Final state (F)
5. Stack Alphabet (Γ)
6. stack symbol (Z_0)
7. Transition symbol (δ)

(27) ~~How to~~ Two Languages accepted by CFG.

i) Accepted by final state.

ii) Accepted by empty stack

(28) Theory

(29) PDA has stack.
NFA doesnot have stack.

(30) NPDA accepts large language than DPDA. So NPDA is superior.

(31) NPDA & DPDA are not same.
NPDA accepts larger language than DPDA

(32) Instantaneous description of PDA

If $P = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, F\}$.

$\delta(q_0, X_0, q_0) = (q_1, X\beta)$.

(33) Defing a language in PDA using two methods.

i) languages having well formed parantheis

ii) $L = \{a^n b^n \mid n \geq 1\}$.

iii) $L = \{a^n b^m \mid n \geq 1, m \geq 1\}$.

(34) Doubt

(35)

NFA	PDA
Non deterministic Finite Automata.	Push Down Automata
It does not have memory.	It has memory
There is no stack	There is stack.
Two types ⇒ NFA ε NFA.	Two types ⇒ DPDA NPDA.

(36) PDA has memory?

* PDA has memory, it accepts LFG.
* It can operate on palindrom $a^n b^n$, $a^n b a^n$

(37) DPDA:

when the transition leads to any one transition it is DPDA

(38) Diff b/w final state method & stack method.

* In final state method, the PDA enters final state on reading a valid string.
* In stack method, the stack empties on reading the valid string.